HL7 AI COMPETITION

WSO2

Care Loop

Supporting evidence for the solution document: the clinical basis, the FHIR and AI implementation, and the security and safety detail behind the claims.

SUPPORTING EVIDENCE

COMPANION TO	STANDARD	DATE
Solution Document	HL7 FHIR R4	July 2026

SUPPORTING EVIDENCE

The clinical, technical, and safety evidence behind the Care Loop.

This document backs the claims in the solution document with detail: the clinical literature the design rests on, the exact FHIR resources the platform produces and consumes with worked examples, a card for the machine-learning model, the AI agent specification, and the security, observability, and safety posture. It is written for a reader who wants to check the work.

CONTENTS

A Clinical evidence base

Each design decision in the Care Loop traces back to the literature. The table maps the evidence to the claim it supports.

FINDING	SOURCE	WHAT IT SUPPORTS IN THE CARE LOOP
Wearable data flags HF deterioration a median of 7.4 days early; each 10% drop in wearable-derived daily pVO_2 carries a 3.62× hazard for unplanned events (Pearson 0.85 vs. clinic CPET).	Remote monitoring of heart failure exacerbations using a smartwatch. Nature Medicine 32:924–933, 2026.	The core mechanism: a consumer wearable, read continuously, buys days of actionable warning.
~64 million people live with HF; ~US$346 billion annual global cost; median survival 1.7 y (men) / 3.2 y (women).	Global Public Health Burden of Heart Failure review; Nature Medicine 2026 introduction.	Scale and cost of the problem the platform addresses.
HFrEF five-year mortality 43–75%.	Burger et al., Eur. J. Heart Failure, 2023.	Why HFrEF specifically is the target population.
~25% five-year survival after an HF hospitalisation.	HFrEF: A Review, JAMA, 2020.	The stakes of each avoidable admission.
~1 in 5 readmitted within 30 days; ~1 in 3 within a year.	Khan et al., Circ. Heart Failure; Global Comparison of Readmission Rates, JACC, 2023.	The revolving-door cost early intervention targets.
~2-week silent deterioration window before symptoms appear.	SCALE-HF 1, J. Cardiac Failure, 2024.	The window the platform acts inside.
Remote monitoring lowers mortality and hospitalisation in RCTs.	De Lathauwer et al., Eur. J. Heart Failure, 2025; TIM-HF2, Lancet, 2018.	Evidence that acting on the signal changes outcomes.

B FHIR resource catalogue and examples

The platform is FHIR R4 end to end, served by a conformant FHIR server at the EHR and a second instance as the Care Loop's own store. The two are kept in step by a sync service that mirrors resources from the EHR into the internal store hourly, walking them in dependency order (Patient, then Encounter / Condition / MedicationRequest / AllergyIntolerance, then Observation) so references resolve. Terminologies used: LOINC (vitals), UCUM (units), SNOMED CT (conditions, allergies), RxNorm (medications), and HL7 code systems for observation category and risk probability. Patient consent is currently a clinic enrolment step; capturing it as a structured, revocable FHIR `Consent` resource is design intent, noted here for completeness.

B.1 VITALS AS AN OBSERVATION (LOINC + UCUM)

Each hourly reading becomes a vital-signs Observation, delivered inside a transaction Bundle. Heart rate 8867-4, SpO_2 59408-5, respiratory rate 9279-1, systolic BP 8480-6, diastolic BP 8462-4.

```
{
  "resourceType": "Observation",
  "status": "final",
  "category": [{ "coding": [{ "system": "http://terminology.hl7.org/CodeSystem/observation-
category",
    "code": "vital-signs" }] }],
  "code": { "coding": [{ "system": "http://loinc.org", "code": "8867-4", "display": "Heart rate"
}] },
  "subject": { "reference": "Patient/patient-123" },
  "effectiveDateTime": "2026-07-15T09:00:00Z",
  "valueQuantity": { "value": 98, "unit": "beats/minute",
    "system": "http://unitsofmeasure.org", "code": "/min" }
}
```

B.2 THE ESCALATION TASK HANDED TO THE CLINIC

When both signals agree, the analysis service writes a Task into the EHR, prioritised STAT / URGENT / ROUTINE by the worse of the two probabilities, with `basedOn` references to the evidence so a clinician can trace the "why".

```
{
  "resourceType": "Task", "status": "requested", "intent": "order",
  "priority": "urgent",
  "code": { "text": "Review remote monitoring alert" },
  "for": { "reference": "Patient/patient-123" },
  "authoredOn": "2026-07-15T09:04:00Z",
  "basedOn": [
    { "reference": "RiskAssessment/risk-ml-001" },
    { "reference": "RiskAssessment/risk-agentic-001" },
    { "reference": "QuestionnaireResponse/symptom-001" }
  ]
}
```

The description text is written by a dedicated agent for two readers at once: a front-desk operator and a clinician.

B.3 THE AGENTIC RISKASSESSMENT

The reasoning agent's judgement is stored as a RiskAssessment carrying its probability, an HL7-coded qualitative risk level (`.../risk-probability`), its short reasoning as a note, and the specific resources it cited as its basis.

FIELD	CONTENTS
status	`final`
prediction.probabilityDecimal	The agent's own probability of a cardiac event (0–1).
prediction.qualitativeRisk	low / moderate / high, coded against the HL7 risk-probability system.

FIELD	CONTENTS
note.text	The agent's 2–3 sentence reasoning, naming the values it relied on.
basis	References to the Observation / Condition / MedicationRequest resources cited, by real id.

C The AI: agents, model, and gateway

The agents are built with Ballerina's native AI support (`ai:Agent` with an `ai:McpToolKit`). Three purpose-built agents split the work, each reaching the patient's live FHIR record through the shared FHIR MCP tool.

AGENT	MODEL	ROLE AND GUARDRAILS
Questionnaire	GPT-4.1 nano	Drafts 4–6 plain-language symptom questions matched to the recent vitals trend as a FHIR Questionnaire (no answers), delivered as chat; only the patient's QuestionnaireResponse is persisted.
Risk assessment	GPT-4.1	Reasons over vitals, active conditions, medications, and history; excludes resolved conditions; cites at most five resources by real id; assesses and reports only, never decides to escalate.
Task description	GPT-4.1 nano	Writes the case narrative for the front desk and clinician from what it was given; invents nothing.

> THE DECISION IS NOT THE AGENT'S
>
> The escalation decision is a deterministic rule outside the agents: a case is raised only if the ML probability and the agentic probability both clear their thresholds (each 0.5 in the reference build). Task priority is then set by the worse of the two probabilities: ≥0.85 STAT, ≥0.65 URGENT, otherwise ROUTINE. The AI narrows and explains; the rule decides.

Every LLM call is routed through the **WSO2 Agent Manager AI gateway** (AMP), so the model provider is a configuration choice and each call is authenticated, rate-limited, and traced. The agents read FHIR through the MCP bridge directly; they never call the gateway to reach clinical data, and they never write it.

D Machine-learning model card

The hourly risk score comes from a small, honest model over the few signals a watch and a health profile can actually supply. It is deliberately not a black box promising more than its inputs support.

Task	Produce an hourly risk score from a patient's profile and heart-rate signal. The label is a risk proxy learned from a public diagnostic dataset, not a validated predictor of a heart-failure exacerbation event.

Model	Gradient-boosted trees (XGBoost), selected by cross-validated ROC-AUC from a 10-model search, exported to ONNX.
Inputs	Age, maximum heart rate, sex — the three signals a watch plus a HealthKit profile can supply. The other collected vitals (SpO_2, respiratory rate, BP) are stored as FHIR but are not model inputs. Preprocessing (scaling, encoding) is baked into the ONNX graph.
Output	A score in [0,1] for the positive class, plus a thresholded label (default 0.5).
Training data	A public heart dataset (734 train / 184 test).
Held-out performance	ROC-AUC 0.80, accuracy 0.76, precision 0.80, recall 0.75, F1 0.78. This measures the scoring capability on the dataset's own target, not deterioration-prediction performance.
Runtime	ONNX Runtime (CPU), no training stack or GPU required; portable across CPU and GPU.
Honest limits	Trained on a public dataset, not a purpose-built HFrEF cohort; a fielded model would need prospective validation and regulatory work. See solution document, Section 9.

> The model is intentionally the cheap, constant layer: it runs every hour so a full day of hourly scores already exists to reason over, and it never calls an LLM. The expensive agentic reasoning is reserved for the hours it escalates.

E Security, safety, and observability

- **Read-only AI by construction.** The agents reach clinical data only through the FHIR MCP bridge, which sits behind a proxy that refuses anything but GET/HEAD. An LLM physically cannot mutate a patient record, regardless of prompt. The one write-back, the Task, is performed by the deterministic analysis service.
- **Two independent signals to escalate.** A statistical model and a reasoning agent must both cross their thresholds before any case reaches a person, guarding against a single noisy reading or an over-confident agent.
- **Fail safe, not silent.** If an urgent check-in goes unanswered within the timeout window (4 hours in the reference build), the case escalates on the model's signal alone rather than being dropped — silence on an emergency question is treated as a bad sign. The counterpart risk, a deterioration that never crosses the model threshold and so is never checked in on, is why the platform is positioned as a prioritisation aid and not a substitute for a patient seeking care.
- **Grounded, auditable reasoning.** The risk agent must cite the specific FHIR resources supporting its conclusion by real id and may never invent one, so a clinician can check the evidence rather than trust the model.
- **The LLM boundary is explicit.** The reasoning agents send clinical data to an LLM, so PHI leaves the provider boundary when a hosted model is used. The AI gateway makes the provider a configuration choice, so a provider can run a self-hosted or in-jurisdiction model to keep PHI inside

their boundary; a hosted provider requires a business-associate / data-processing agreement and minimum-necessary data. This is how the self-hostable claim holds against HIPAA / GDPR rather than contradicting it.

- **Observable AI, and access to be audited.** Every LLM call is traced through the AI gateway; spans are exported over OpenTelemetry and scoped to the registered agent, giving a provider a record of what the AI saw and did. Human access to the dashboards sits behind the provider's authentication, with AuditEvent-style logging of PHI access as the expected, not-yet-built, control.

F Deployment and component inventory

The stack is container-deployed and runs from a single command for evaluation. Integrations are written in Ballerina; the ML service in Python; the dashboards and simulators in TypeScript.

COMPONENT	STACK	ROLE
Apple HealthKit simulator	Python / FastAPI	Produces hourly vitals as FHIR Observation bundles.
Messaging simulator	TypeScript / Next.js	Delivers the questionnaire and captures the patient's answers.
Collector service	Ballerina	Ingests vitals, opens check-ins, builds QuestionnaireResponses.
Heart-risk service	Python / ONNX Runtime	Serves the ML probability from the vitals.
Analysis service	Ballerina	Holds the decision logic: thresholds, escalation, the Task write-back, the timeout fail-safe.
AI service	Ballerina AI	The three agents, routed through the AMP AI gateway.
FHIR servers (×2)	WSO2 FHIR R4	The EHR source of truth and the Care Loop's synced internal store.
FHIR MCP bridge + proxy	WSO2 / Nginx	Read-only FHIR-to-MCP access for the agents.
Front-desk & clinician dashboards	TypeScript / Next.js	Triage and review of the raised cases.
Agent Manager + OTel collector	WSO2 AMP	AI gateway, agent identity, and distributed tracing of LLM calls.

> The reference deployment uses simulators for the patient-side transports; the FHIR they produce and the loop they drive are real. Substituting the live HealthKit webhook and a real messaging integration does not change the rest of the system.